Leveraging Hierarchical Semantic Maps for Natural Language Communication

Angel Daruna[1] and **Sonia Chernova**[2]

***Abstract*— Semantic maps serve a multifaceted purpose for robots in HRI. To name a few aspects, robots can leverage these maps for plan optimization, natural language interpretation, and natural language expression. This work is focused on the later of these three aspects. Specifically, we represent the semantic map as a tree of semantic volumes. This map is used to express the locations of various points in the environment using unequivocal natural language. These points include the location of the robot, the locations of items, and the location of the robot's center view point. Various topological structure cases are studied for natural language communication of ambiguous point locations. For the most complex case, where a set of ambiguous labels has no unique ancestors in the semantic map, the coefficient of variance was used to allow the robot to decide which feature to mention that would disambiguate the natural language location.**

I. INTRODUCTION

A semantic map is a state representation internal to the robot that allows the robot to associate spatial points with semantic labels, such as *kitchen* or *table*. Semantic maps are often implemented as hybrid maps [1] that link separate metric and topological maps. The metric map is used to measure the spatial location of points in the environment. The topological map is used to represent human labels about the environment. The two maps are then combined using links in order to relate human labels to the spatial locations of points. These links can be as simple as associating a volume in the environment with a semantic label.

The semantic labels can relate to other knowledge the robot has access to [2] or may simply be a human domain term [3]. In either case, these semantic labels allow for more meaningful human-robot interactions. They endow the robot with the ability to reason about the environment relative to a human's cognitive representation [4], interpret natural language about tasks [5], generate natural language about their internal states, and perform tasks using semantic labels [6]. This preliminary study develops a semantic map and uses the semantic map to communicate unambiguous robot belief states.

The rest of this document is organized as follows. Section II presents similar works, Section III explains the semantic map implementation. Section IV goes over various topological structure cases that cause semantic localization ambiguity and the solution proposed to make natural language locations unambiguous. Section V shows some responses generated by the robotic platform used in lab, Stanley Innovation's Vector robot. Finally, Section IV discusses future work.

[1]A. Daruna is a doctoral student in the Electrical and Computer Engineering Dept. (Georgia Institute of Technology) `adaruna3 at gatech.edu`

[2]S. Chernova is the Catherine M. and James E. Allchin Early-Career Assistant Professor in the School of Interactive Computing (Georgia Institute of Technology) `chernova at cc.gatech.edu`

II. RELATED WORK

Much of the earlier work in semantic mapping focused on the autonomous construction of semantic maps. Galindo et al. showed how a hybrid hierarchical semantic map could be created by the robot as it navigates indoor environments. The map was a hybrid map that connected separate topological spatial and conceptual maps via anchors. These anchors connected abstract concepts in a knowledge base, like *living room,* to objects that were recognized in the environment. This allowed conceptual information in the knowledge base to be leveraged for semantic mapping. More recent work in semantic map building has gone further to use heterogeneous perceptions to build probabilistic semantic maps [7]. They used different levels of abstraction to combine the multi-modal sensor data into a common conceptual representation that could then be used to reason about the likelihood of a room type or objects being located in the room.

Aside map building, previous works have also explored the advantages of using semantic maps. Galindo et al. showed the use of semantic maps improves the performance of their planning algorithms. Madrigal et al. used a multi-hierarchy semantic map to improve the autonomous capabilities of an advanced wheelchair robot. The hierarchies allowed the robot to navigate environments with complex topologies given a set of instructions, like *"Go to Library"*. A similar work used semantic fields to allow service robots to interpret natural language and complete tasks [5]. These fields allowed for a probabilistic estimate of how closely different objects in the environment matched the natural language instructions in order for the robot to complete the task.

Less attention has been given to the communication of internal robot belief states using semantic maps. A similar study to this work shows how a robot could use spatial language to express its beliefs about the environment [4]; however, this language was limited to objects immediately in the vicinity of the robot that could be expressed with relationships similar to *"The object #3 is in front of me but extends to the right relative to me. The object is very close"*. The contribution of this work is to perform a similar task that gives an unambiguous description of an object location in natural language, but uses a semantic map to express the location relative to the environment rather than relative to the robot.

III. The Semantic Map

The semantic map is implemented as a hybrid map of a metric map and a topological map. According to the taxonomy proposed in [1], this semantic map is of the heterogeneous, hierarchical, and separable kind. The metric map is used to measure points in the environment relative to a global frame. The topological map captures human domain knowledge about the environment in a tree graph. The links between the topological map and metric map are expressed directly in the topological map nodes. Each node is associated with a certain volume located in the metric map. This means that any points in the node's volume are also in the semantic label. Therefore, it is straight forward to know whether a point is in a *'kitchen'* for example. Furthermore, because the topological map is hierarchical, it is just as simple to know if a point is within a drawer in the kitchen.

A semantic map was created for the lab where the robot was tested. Below are the topological and metric maps that make up the semantic map on the right in Fig. 3. Because the focus of this work is the unequivocal communication of the locations of points within the environment, the semantic volumes are assumed to be given using a method similar to previous works. This may be an avenue for future work.

Fig. 1. Metric map with an overlay of the topological node volumes

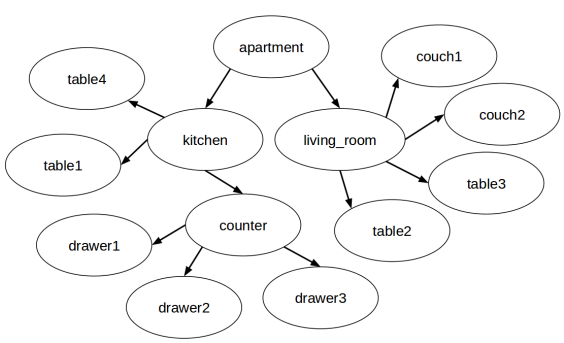

Fig. 2. Topological map showing semantic labels

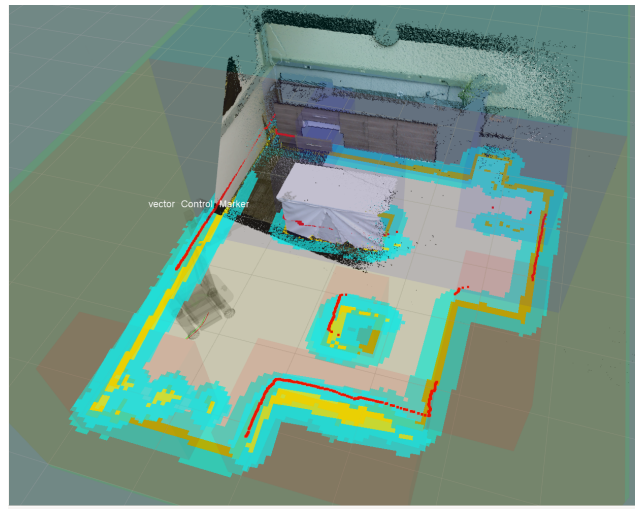

Fig. 3. Semantic Map implementation in rviz

IV. Semantic Label Ambiguity

The semantic labels presented in the topological map are the original labels used by the robot shown in Fig. 2. These are unambiguous because each instance of an object has a unique label (drawer1, drawer2; couch1, couch2). However, the labels are not convenient for natural language communication for obvious reasons. Therefore, the labels should be stripped of their unique instance identifier so they can be used in natural language. This immediately leads to ambiguities in human-robot interaction through natural language.

This relabeling leads to three main cases that need to be considered for disambiguation.

- **Unique label:** the stripped label is still unique. This is the case for *counter* in the topological map of Fig. 2.
- **Unique ancestor:** the stripped label is not unique, but it's parent or ancestor has a stripped label that is unique. This is the case for *table1* and *table2* in the topological map of Fig. 2.
- **Unique attribute:** the stripped label is not unique, and contains no unique ancestors. A unique attribute about the node volume must be used to disambiguate the labels. This is the case for *table1* and *table4* in the topological map of Fig. 2.

The first two cases have straight forward solutions. For the first case of unique label, the solution is given by the label. For the second case of unique ancestor, the solution is given by combining the ambiguous label with a label of the unique ancestor. This can be done using an *in the* relation because of the hierarchical structure of the semantic map.

For the final case, no unique label in the tree exists. Therefore, we must use the node volumes to disambiguate a set of ambiguous labels. This can be done using an arbitrary set of attributes about the node volumes, like the length of the volume, the size of the volume, the color of the item contained in the volume, and so on. These attributes can then be used to disambiguate the set of semantic labels by choosing

the attribute that has the highest variance between the set. For the purposes of this experiment, height and volume attributes were used. Because measuring the variance directly would bias the attributes being selected, a dimensionless value that is still meaningful for a small number of values is used to compare the variances of different attributes, such as the coefficient of variance,

$$CV = \frac{\mu}{\sigma},$$

which is the mean μ divided by the standard deviation σ.

V. Natural Language Locations

Table 1 shows the natural language responses the robot gave for various query points. The robot automatically decides the necessary information needed in the natural language response to express the unequivocal natural language location. In addition to the robot's view point (i.e. the center of the field of view as measured by averaging several points in the middle of a point cloud) and the robot's location, you can also query for the natural language locations of any arbitrary points in the semantic map.

TABLE I

Robot responses to various point queries

Actual Semantic Location in Fig. 2	Queried Point	Response
living_room	Robot Location	'I am in the living_room'
table2	Robot View Point	'I am looking at the 2nd largest table in the living_room'
kitchen	Robot Location	'I am in the kitchen'
drawer2	Robot View Point	'I am looking at the 2nd highest table in the kitchen'

VI. Conclusions

This preliminary work has show that the coefficient of variance is can be reliably used to compare variance of disparate object attributes. While this work does allow the robot to be queried for arbitrary points in the environment, it does not yet account for dynamic objects in the environment. It would be a simple extension to allow the topographical map to automatically include new and remove old edges that connect nodes associated with dynamic volumes using object recognition. This would allow the robot to track and express the semantic locations of objects as they move in the environment.

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